PREDICTED PROTEIN RECORD

AT1G49530.1

Candidate retained by the two-filter homology and Pfam architecture workflow, with Evo2 retained as independent model validation.

Predicted protein | Arabidopsis thaliana | Genome Ath1 | EC 2.5.1.29 | 336 aa | 1011 nt | High Pfam confidence

HIGH
PFAM

ARCHITECT
ExactMa

SUBJECT ID	SPECIES	GENOME	EC NUMBER	IDENTITY	QUERY COVERAGE
AT1G49530.1	Arabidopsis thaliana	Ath1	2.5.1.29	61.446%	86.0%

HOMOLOGY EVIDENCE

BLASTp evidence

Primary sequence-similarity evidence for this candidate protein.

IDENTITY	COVERAGE	E-VALUE
61.446%	86.0%	1.51e-134
Sequence similarity	Aligned query coverage	BLAST significance

FUNCTIONAL ARCHITECTURE

Domain architecture

Pfam domain positions detected by local HMMER search.

Protein length: 336 aa

1 confident hit · 2 weak / filtered hits



Confident domain Weak / filtered domain

CONFIDENT DOMAIN HITS 1

DOMAIN	PFAM	START	END	I-EVALUE	SCORE
polyprenyl_synt	PF00348.23	67	307	3.80e-56	190.7

WEAK / FILTERED DOMAIN HITS 2

DOMAIN	PFAM	START	END	I-EVALUE	SCORE
Mre11_C_bact	PF22161.2	204	217	9.30e+01	3.7
Mre11_C_bact	PF22161.2	258	272	2.40e+01	5.6

NAV
On
BLA
Dom
PLM
Evo2
Sequ

ADDITIONAL VALIDATION

Protein language model-based homology evidence

ProstT5-predicted 3Di sequences compared with curated enzymes using Foldseek.

HOMOLOGY CONFIDENCE	BEST-MATCHED CURATED REFERENCE FOR THIS REACTION STEP	FOLDSEEK E-VALUE	QUERY COVERAGE	TARGET COVERAGE	BIDIRECTIONAL COVERAGE
High	P34802	4.19e-31	97.3%	83.0%	83.0%

High-confidence protein-language homology evidence against this curated reference; coverage values report the aligned extent.

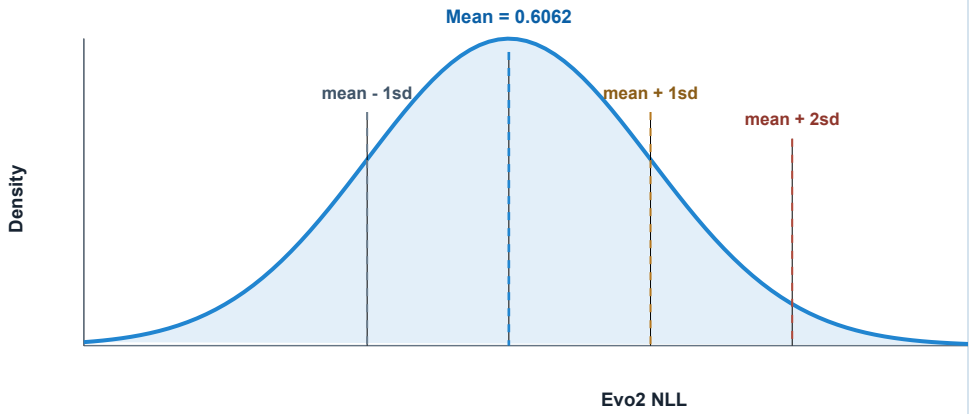
▸ Confidence criteria

AUXILIARY MODEL VALIDATION

Evo2 assessment

Independent DNA language-model evidence for candidates retained by the two-filter workflow.

CURRENT NLL	KNOWN MEAN NLL	KNOWN STD NLL	Z-SCORE
0.8748	0.6062	0.0687	3.9118



Evo2-supported
Z-score <= 1

Evo2-borderline
1 < Z-score <= 2

Evo2-weak
Z-score > 2

Current status
Evo2-weak

NLL values generated by the Evo2 DNA language model are compared with known enzymes in the same pathway context. This section is auxiliary validation and does not replace the homology and Pfam architecture filters.

PRIMARY RECORD

Sequence record

Protein and CDS sequences retained for inspection and downstream analysis.

Protein | CDS

Copy sequence

336 amino acids

MRPRYSLILSAMRLIRPSNRRLSSIASSDSEFISYMKNKAKSINKALDNSIPLCNNFVPLWEPVLEVHKAMRYTLLPGGKRVRPMLCLVACELVGGQESTAMPAACA
VEMIHAASLILDDLPCMDDDSLRRGKPTNHKVFGEKTSILASNALRSLAVKQTLASTSLGVTSERVLRAVQEMARAVGTEGLVAGQAADLAGERMSFKNEDDELR
YLELMHVHKTAVLVEAAAVVGAIMGGGSDEEIERLKSYARCVGLMFQVMDDVLDETKSSEELGKTAGKDLITGKLTYPKVMGVDNAREYAKRLNREAQEHLQGFD
SDKVVPLLSLADYIVKRQN

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